Some Fundamentals of B. F. Skinner's Behaviorism

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Despite B. F. Skinner's prominence, his impressive written corpus, and the many authoritative presentations by others of his approach to psychology, the fundamentals of Skinner's psychology have never been addressed in any comprehensive manner. In this article, the authors take steps to fill this gap by synopsizing Skinner's written corpus into 12 fundamental points that seem to characterize his behaviorism.

Behaviorism's impact on disciplines inside and outside of psychology is exemplified by the prominence of its leading advocate for much of this century, B. F. Skinner (Gilgen, 1982; Heyduk & Fenigstein, 1984). Skinner's version of behaviorism continues to exert a significant influence on psychology and the culture at large. Reviewers who have conducted quantitative (Wyatt, Hawkins, & Davis, 1986) and qualitative assessments (Leahey, 1987) agree that Skinner's psychology is alive and well. A random sample of members of the American Psychological Association ranked Skinner first in a survey of the most important people in American psychology during the post-World War II period (Gilgen, 1982). In another survey of the most important events and influences in post-World War II American psychology, a sample from the same source ranked Skinner's contributions first, behavior modification (largely associated with Skinner) second, and the growth of behavioral psychology fourth (Gilgen, 1982). Thus, it is an understatement to conclude that Skinner has been, and is, influential and well-known.

Despite Skinner's influence, his impressive written corpus, and the many authoritative and comprehensive presentations by others of his approach to psychology (e.g., Catania, 1980; Michael, 1985; Reese, 1986), no one has forthrightly addressed the fundamental features, including assumptions, of Skinner's approach to psychology (but see Nye, 1979; Skinner, 1974; Verplanck, 1954, for some preliminary attempts). Given Skinner's influence and scholarship, this strikes us as an oversight. In an attempt to fill this gap, we present what we consider to be 12 fundamental points of Skinner's behaviorism.

In this presentation, we adhere to a format that includes a concise statement for each assumption, followed by at least two quotations from which it was derived. This is followed by a discussion of each. The quotations used are those that, after a careful analysis of Skinner's published works, seemed to best represent his position on particular issues. Although we have attempted to minimize interpretations and translations of these quotations, we cannot be certain that a sufficient amount of context was taken into account when drawing conclusions from the written data. It is possible that variations in context would have led to modifications of at least some of the features on the basis that the quotations we used are not representative samples. Arguing for their veracity is the relatively high degree of internal consistency in Skinner's overall system as we present it.

The features we identified are organized in a quasi-logical order such that those presented later build on those presented earlier. This organizational scheme reflects our own way of synthesizing Skinner's psychology into a coherent whole. We do not critically assess the features either singly or in toto. Our goal has been to synopsize the psychology of the most eminent psychologist of the latter part of the 20th century. The points we address pertain to the purpose of science, methodology, determinism, locus of behavioral control, consequential causality, materialism, behavior as subject matter, reductionism, nonreductionism, organism as the locus of biological change, classification of behavior into respondent and operant, stimulus control of operant behavior, and the generality of behavioral principles.

Purpose of Science: The Primary Purpose of Science is Prediction and Control

We undertake to predict and control the behavior of the individual organism. (Skinner, 1953, p. 35)

The object [of my research] has been to discover the functional relations which prevail between measurable aspects of behavior and various conditions and events in the life of the organism. The success of such a venture is gauged by the extent to which behavior can, as a result of the relationships discovered, actually be predicted and controlled. (Skinner, 1972, pp. 257–258)

If we have achieved a true scientific understanding of man, we should be able to prove this in the actual prediction and control of his behavior. (Skinner, 1972, p. 259)

The laboratory techniques . . . and their technological applications, emphasize the prediction and control of behavior via the manipulation of variables. Validation is found primarily in

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the success with which the subject matter can be controlled. (Skinner, 1972, p. 41)

Skinner offered prediction and control as the primary goals of science instead of hypothesis or theory testing. He opposed deductive methods, which purported to postulate a theory a priori and then test it against empirical evidence. Skinner obtained empirical data first and then, by induction, derived general principles or functional relations between events. To ensure that the relations thus described actually pertain to the events investigated, he suggested that the scientist use them to make predictions and to control subsequent events. Once the events are successfully predicted and controlled, the relations discovered are confirmed.

Skinner's emphasis on prediction and control over theory and hypothesis testing directly relates to a much misrepresented aspect of his systematic position. Although he replaced theory testing with prediction and control, he only abjured conventional psychological theorizing. In his article "Are Theories of Learning Necessary?" Skinner (1950) described the class of theory he rejected as "any explanation of an observed fact which appeals to events taking place somewhere else, at some other level of observation, described in different terms, and measured, if at all, in different dimensions" (p. 193). However, Skinner vigorously stressed the importance of a theory of behavior:

Behavior can only be satisfactorily understood by going beyond the facts themselves. What is needed is a theory of behavior, but the term "theory" is in such bad repute that I hasten to explain [that not needed are theories of the conventional type]. (1947, pp. 27–28)

Whether particular experimental psychologists like it or not, experimental psychology is properly and inevitably committed to the construction of a theory of behavior. A theory is essential to the scientific understanding of behavior as a subject matter. (1947, pp. 28–29)

Beyond the collection of uniform relationships lies the need for a formal representation of the data reduced to a minimum number of terms. A theoretical construction may yield greater generality than any assemblage of facts. (1950, pp. 215–216)

Skinner (1947) outlined three basic steps for constructing a theory: (a) Decide on the basic data (the events we seek to understand), (b) collect data (functional relations, facts), and (c) inductively develop explanatory (theoretical) concepts. The addition of facts permits the emergence of collections of concepts (i.e., theory). The following statement shows Skinner's (1947) affirmation of theorizing and the central role of prediction and control in this process:

We need to arrive at a theory of human behavior which is not only plausible, not only sufficiently convincing to be "sold" to the public at large, but a theory which has proved its worth in scientific productivity. It must enable us, not only to talk about the problems of the world, but to do something about them, to achieve the sort of control which it is the business of a science of behavior to investigate. The superiority of such a theory will then be clear and we shall not need to worry about its acceptance. (p. 46)

Skinner's view that the essence of scientific behavior is prediction and control comported with his position on the epistemological question of the nature of scientific knowledge. In the midst of discussions of operationism in psychology, Skinner (1945a) argued against intersubjective agreement as the major criterion for the acceptance of scientific knowledge. He suggested that "whole-hearted agreement on the definition of psychological terms . . . makes for contentment but not for progress" (1945b, p. 293). As he put it:

The ultimate criterion for the goodness of a concept is not whether two people are brought into agreement but whether the scientist who uses the concept can operate successfully upon his material—all by himself if need be. What matters to Robinson Crusoe is not whether he is agreeing with himself but whether he is getting anywhere with his control over nature. (1945b, p. 293)

Thus, Crusoe's concern is prediction and control. Skinner elaborated on this pragmatic theory of truth (cf. Zuriff, 1980) numerous times; for example:

Knowledge enables the individual to react successfully to the world about him just because it is the very behavior with which he does so. (1953, p. 409)

[Scientific knowledge] is a corpus of rules for effective action, and there is a special sense in which it could be "true" if it yields the most effective action possible. (1974, p. 235)

A proposition is "true" to the extent that with its help the listener responds effectively to the situation it describes. (1974, p. 235)

Skinner's pragmatic epistemology carried over to how he approached relations between science and technology and to his concern for the culture at large. He took the position that a technology most needs principles for effective action and, in a like manner, that the very survival of a culture depends on successful control over conditions that threaten it (Skinner, 1971, 1978, 1987a, 1989). Skinner frequently argued that science based on prediction and control was preferable to one founded on theory testing when we seek to address applied (extralaboratory) problems. Furthermore, Skinner's position on the purpose of science and his pragmatic epistemology directly related to his practice of using individual-organism experimental design tactics. He suggested that "no one goes to the circus to see the average dog jump through a hoop significantly oftener than untrained dogs raised under the same circumstances" (Skinner, 1956, p. 228).

Methodology: The Methodology Is Functional Analysis, Which Relates Environmental Independent Variables to Behavioral Dependent Variables

[Experimentation means] we manipulate certain "independent variables" and observe the effect upon a "dependent variable." In psychology the dependent variable, to which we look for an

effect, is behavior. We acquire control over it through the independent variables. The latter, the variables which we manipulate, are found in the environment. (Skinner, 1947, p. 20)

We undertake to predict and control the behavior of the individual organism. This is our "dependent variable"—the effect for which we are to find the cause. Our "independent variables"—the causes of behavior—are the external conditions of which behavior is a function. Relations between the two . . . are the laws of a science. (Skinner, 1953, p. 35)

Skinner's interest was psychology as an experimental science. Experimentation allows the researcher to identify reliable relations between one class of variable—the manipulated environmental class—and behavioral ones. Skinner labeled as *functional relations* those relations that occur when a change in an independent variable results in a change in a dependent variable. The process of experimentation leading to the identification of functional relations was labeled *functional analysis.* Functional analysis yields functional relations that are the basic facts of the science of behavior.

The key point of Skinner's methodology is the connection among experimental functional analysis, functional relations, and what he meant by *controlling variables* (used synonymously with conditions under which behavior occurs and with conditions, or variables, of which behavior is a function). The independent variables of functional relations are the controlling variables that permit the scientist to predict and control behavior. Skinner (1953, pp. 32–33) illustrated this process using the example of predicting and controlling drinking a glass of water. We can control drinking by manipulating variables such as deprivation history, room temperature, exercise, and amount of salt or urea in food ingested before the experiment. To predict whether or not the subject will drink, we must have information on each of these controlling variables and on "extraneous" ones as well. Accordingly, appeal to hypothetical states or conditions (e.g., motivation, drive, thirst, feelings) thought to be induced by independent variables as causal variables is to propose explanatory fictions that forestall scientific understanding because they "allay curiosity and . . . bring inquiry to an end" (Skinner, 1957, p. 6). That is, the search for controlling variables that lie outside the organism is truncated when, for example, we say the person perspired and stuttered because of anxiety instead of searching for environmental variables that control the excessive perspiration and disfluent speech.

Classic applications of experimental methodology are based on the assumption that it provides identification of cause-and-effect relations; however, Skinner (1953) departed from a strict adherence to this aspect of experimentation:

A "cause" becomes a "change in an independent variable" and an "effect" a "change in a dependent variable." The old "cause-and-effect connection" becomes a "functional relation." The new terms do not suggest *how* a cause causes its effect; they merely assert that different events tend to occur together in a certain order. This is important, but it is not crucial. There is no particular danger in using "cause" and "effect" in an informal discussion if we are always ready to substitute their more exact counterparts. (p. 23)

A somewhat subtle aspect of Skinner's methodology is that the independent variables of most interest are selective contingencies (see Consequential Causality Section) to which the organism was exposed before the occurrence of the instance of the behavior to be explained. This view of independent variables as temporally remote in the past of the organism departs from conventional applications of the experimental model in which "causal" variables are required to be immediately antecedent to effects, sometimes requiring hypothesized mental causes. An experimental methodology with (causal) independent variables whose effects are detected after a period of time contrasts with the conventional behavioristic view of experimentation in which independent and dependent variables refer to temporally contiguous stimuli and responses, respectively.

Determinism: Behavior Is Determined; It Is Lawful

[Science] is more than the mere description of events as they occur. It is an attempt to discover order, to show that certain events stand in lawful relations to other events. . . . If we are to use the methods of science in the field of human affairs, we must assume that behavior is lawful and determined. (Skinner, 1953, p. 6)

To have a science of psychology at all, we must adopt the fundamental postulate that human behavior is a lawful datum, that it is undisturbed by the capricious acts of any free agent—in other words, that it is completely determined. (Skinner, 1947, p. 23)

I was working on a basic Assumption—that there was order in behavior if I could only discover it—but such an assumption is not to be confused with the hypotheses of deductive theory. (Skinner, 1956, p. 227)

Man is not made into a machine by analyzing his behavior in mechanical terms. Early theories of behavior . . . represented man as a push-pull automaton, close to the nineteenth-century notion of a machine, but progress has been made. Man is a machine in the sense that he is a complex system behaving in lawful ways, but the complexity is extraordinary. (Skinner, 1971, p. 202)

[In reference to inconspicuous determining events and conditions that are "easily overlooked"]: It is then easy to believe that the will is free and that the person is free to choose. The issue is determinism. The spontaneous generation of behavior has reached the same stage as the spontaneous generation of maggots and micro-organisms in Pasteur's day. (Skinner, 1974, pp. 53–54)

Skinner followed the commonly accepted position that the scientific method begins with a deterministic assumption rather than an indeterministic one. The scientist assumes lawfulness, hence determinism, and proceeds to look for lawful relations. Skinner was not different from other early pioneers, such as Freud, who attempted to bring human behavior into the realm of science by adopt-

ing the working assumption that it is orderly and that regularities are able to be discovered by appropriate methods.

In Skinner's approach, this determinism assumption is fundamental for (a) making human behavior amenable to scientific understanding and (b) what Skinner viewed as the primary goals of science: prediction and control. This assumption, however, does not imply any sort of mechanistic determinism in which stimuli and responses are contiguous and the former impel the latter (see Consequential Causality). Indeed, in Skinner's (1935) early work, he described a behavioral relation as the correlation between a stimulus class and a response class, or what might be described today as a definitely molar perspective.

To hold that behavior is determined (i.e., is not capricious) is to hold that it is controlled whether we recognize the lawfulness and sources of control or not. A major point of *Beyond Freedom and Dignity* (Skinner, 1971) is that the most dangerous forms of control are inconspicuous, thus permitting the controlled individual to feel free. Skinner argued that individuals are better off to shed the idea that they are beyond the bounds of controlling factors. He advised individuals to identify how they are controlled and thereby exercise maximum control over their lives rather than leaving their fate in the hands of others who may not have the individual's best interests as a high priority. This point seems to have been neglected uniformly by critics of Skinner's position on freedom and control.

Locus of Behavioral Control: The Causes of Behavior Are Localized in the Environment

> Initiating causes . . . lie in the environment and . . . remain there. (Skinner, 1988e, p. 73)

> The experimental analysis of behavior goes directly to the antecedent causes in the environment. (Skinner, 1974, p. 30)

> The environment made its first great contribution during the evolution of the species, but it exerts a different kind of effect during the lifetime of the individual, and the combination of the two effects is the behavior we observe at any given time. (Skinner, 1974, p. 17)

> What we have learned from the experimental analysis of behavior suggests that the environment performs the functions previously assigned to feelings and introspectively observed inner states of the organism. (Skinner, 1974, p. 248)

The major difference between Skinner and most psychologists who object to his view of behavior as determined, lawful, and controlled revolves around the locus of determining or controlling variables. Skinner objected to the idea that the critical variables for behavior are inside the behaving organism. His view was that there are two possible sources of behavioral control. The autonomous individual approach leads us to search inside the organism for mental structures and processes. The other choice is to examine the environment of the organism. Skinner (1947) found the former not conducive to a scientific approach in contrast to work inspired by the environmental perspective.

Skinner's insistence on nonmental, environmental determinism was consistent with several other aspects of his system. In terms of prediction and control, he maintained that attempts to predict and control behavior by means of organismic-centered causes had failed and that only environmental variables enable the psychological scientist to meet this fundamental goal. To locate controlling variables in the environment is to locate formally the controlling variables of functional relations.

The reliance on environment enables the psychologist to steer clear of nonbehavioral (i.e., mentalistic) explanatory attempts. Related to this is the definition of behavior that includes "commerce with the outside world" (Skinner, 1938, p. 6). In this way, the outside world, or environment, is always an inherent component of behavior, specification of which enables us to determine what the organism is doing. Behavior cannot be separated from the environmental context in which it occurs. Only when one has identified the critical environmental factors determining a bit of behavior has the behavior been defined.

Skinner objected to claims that private events can be used to explain overt behavior: "Although speculation about what goes on within the organism seems to show a concern for completing a causal chain, in practice it tends to have the opposite effect. Chains are left incomplete" (Skinner, 1972, p. 268). The chains are incomplete because the occurrence of the internal event has not been explained. One must ultimately complete the chain by going to the initiating causes in the environment. A behavioral analysis is said to trace the causal chain of behavior back no further than to:

> The point at which effective action [prediction and control] can be taken. That point is not to be found in the psyche, and the explanatory force of mental life has steadily declined as the promise of the environment has come to be more clearly understood. (Skinner, 1974, p. 210)

In placing the causes of behavior in the environment, Skinner maintained consistency with the assumption of materialism (see later discussion) in that reliance is placed on the physical world as opposed to a nonphysical world such as one of mental structures and processes. In placing controlling factors in the environment, the source of the lawfulness in behavior is made consistent with the total system.

Finally, Skinner's attempt to place the locus of behavioral control in environmental events was not a denial that independent variables may be isolated inside the organism, as when injection of a pharmacological agent places a controlling variable within the organism. He denied only that invented internal structures, states, and processes explain behavioral variability. Nor was his view of the locus of behavioral control correctly taken as an "environmentalism" that rules out genetic factors. For Skinner, the opposite to environmental was not heredity but mentality and the assumption of autonomous hu-

mans. Skinner's approach to causality, taken up next, reflects how Skinner departed from earlier behavioristic versions of environmentalism.

Consequential Causality: Selection by Consequences Is the Primary Causal Mode by Which Environment Determines Outcomes in Living Systems

In certain respects operant reinforcement resembles the natural selection of evolutionary theory. Just as genetic characteristics which arise as mutations are selected or discarded by their consequences, so novel forms of behavior are selected or discarded through reinforcement. (Skinner, 1953, p. 430)

Selection by consequences is a causal mode found only in living things, or in machines made by living things. (Skinner, 1981, p. 501)

As a causal mode, selection by consequences was discovered very late in the history of science. . . . The facts for which it is responsible have been forced into the causal pattern of classical mechanics. (Skinner, 1981, p. 502)

Science has largely subscribed to explanation involving one or another version of mechanical causality. Skinner noted that this influential causality of classical mechanics requires initiating agents such as the stimuli of stimulus–response models. In the case of early behavioristic stimulus–response models, the initiating agent was in the individual's external environment. Not fundamentally different from the stimulus–response models of behaviorism was the Freudian and other internalistic applications of mechanics that placed initiating agents inside the organism.

Skinner departed from mechanical causality, derived in part from study of nonliving things, to a type of causality that was discovered in the study of living systems. He argued that Darwinian natural (environmental) selection represents the type of causality most applicable to biology, psychology, and other life sciences. In contrast to mechanical causality, which requires initiating causes, Skinner argued for consequential causality in psychology. According to the principle (Skinner would say "fact") of selection by environmental consequences, behavior occurs and concrete environmental conditions ensue. The effects (changes in behavior) are usually delayed, sometimes considerably, making it difficult for observers to detect the selective process.

Skinner's emphasis on a new causal mode was based on his work in operant conditioning. In the study and application of operant conditioning in its most elementary form, one makes certain consequences available after responses. Over the years, Skinner and his colleagues built on this procedure a sufficiently large body of data to justify the principle most associated with him to the present: Behavior is a function of its (past) environmental consequences. Selection by consequences is a generalization of this principle. In his earliest work, Skinner departed from the antecedent causality of the environmental stimulus–>response model with the operant model that reverses the terms: response–>environment. Thus, it was not difficult for him to raise to the level of a fundamental assumption the principle of consequential causality.

Skinner's stance on causality also led him to a second major departure from environmentalistic hypotheses. His causal mode of selection by consequences applies to both the phylogeny and ontogeny of behavior of species and of individuals, respectively (Skinner, 1966b). Thus, he left behind a bias frequently associated with environmentally oriented theories (i.e., denial or rejection of genetic factors in behavior). Phylogenic contingencies, or contingencies of survival, are behavior–consequence relations that select what appears to be inherited behavior. Ontogenic contingencies, or contingencies of reinforcement, are behavior–consequence relations that select behavior originating during the life of the organism. Because the contingencies responsible for inherited behavior occur in the species' evolutionary history, they are more difficult to confirm than are the reinforcement contingencies that an experimenter can manipulate.

With the extension of selection by consequences from the phylogeny of behavior to the ontogeny of behavior, Skinner counteracted any assertion that his behaviorism is a form of environmentalism that denies heredity. There is clearly a role for genetic participation in behavior through contingencies of survival. However, it is overly simplistic to take genes as determiners of behavior. Skinner's position was that behavior per se is not inherited. This is clarified later in the assumption on the organism as the locus of change (see Organism as the Locus of Biological Change). Instead, susceptibility to ontogenic contingencies is inherited; thus, genetics applies to all behavior just as do those factors commonly referred to as environmental.

Materialism: Dualism Is False; the Only World Is a Physical World

Private and public events have the same kinds of physical dimensions. (Skinner, 1963, p. 953)

The task of a scientific analysis is to explain how the behavior of a person as a physical system is related to the conditions under which the human species evolved and the conditions under which the individual lives. (Skinner, 1971, p. 14)

[My] position can be stated as follows: what is felt or introspectively observed is not some nonphysical world of consciousness, mind, or mental life but the observer's own body. (Skinner, 1974, p. 17)

A small part of the universe is contained within the skin of each of us. There is no reason why it should have any special physical status because it lies within this boundary. (Skinner, 1974, p. 21)

No special kind of mind stuff is assumed. A physical world generates both physical action and the physical conditions within the body to which a person responds when a verbal community arranges the necessary contingencies. (Skinner, 1974, p. 220)

Behaviorists before Skinner attempted to shed dualism. Dualism holds that the world consists of two fundamental

realms: physical (body or matter) and nonphysical (mind or spirit). Matter exists in space and time, whereas mind exists nonspatiotemporally. Most contemporary dualists claim that the physical realm can influence the nonphysical realm, and vice versa (e.g., Eccles, 1989). Skinner held that, because his analyses provided a complete account of behavior without reference to such supposed physical–nonphysical relations, the initial positing of the dualist dichotomy was unnecessary and impedes effective analysis. His alternative is materialism.

Materialism asserts that the world is composed of physical or material things, varying in their states and relations, and nothing else. This monism was clearly expressed by Skinner (1945b): "What is lacking [in dualistic analyses] is the bold and exciting behavioristic hypothesis that what one observes and talks about is always the 'real' or 'physical' world (or at least the 'one' world)" (p. 293). Thus, private events refer to "real" events, and their ontological status is identical to that of any other aspect of the physical world.

Behavior as Subject Matter: The Subject Matter of Psychological Science Is Behavior and Behavior Only

> If psychology is a science of mental life—of the mind, of conscious experience—then it must develop and defend a special methodology, which it has not yet done successfully. If it is, on the other hand, a science of the behavior of organisms, human or otherwise, then it is part of biology, a natural science for which tested and highly successful methods are available. (Skinner, 1963, p. 951)

> Whatever happened to psychology as the science of behavior? (Skinner, 1987b, p. 780)

> Psychology should confine itself to its accessible subject matter [i.e., behavior]. (Skinner, 1987b, p. 785)

For Skinner the first step in developing a theory useful for understanding in a science was to identify the nature of fundamental events. Psychology had to begin with the unequivocal understanding that its events were behavioral. This placed Skinner in the behavioral movement that began in the early 20th century. Taken together with the pragmatic view that prediction and control are the purposes of science, the value and success of any knowledge claim are to be determined by the degree to which the knowledge claimed is useful in predicting and controlling behavior. The alternative to accepting behavior as the subject matter is to opt for mind or mental life, which in turn leads to mental explanations of behavior. Skinner (1963) considered that giving primary emphasis, or any at all, to mind was particularly undesirable because of the failure of this strategy to contribute to the prediction and control of behavior.

We sometimes hear that all of today's psychology is at least partly behavioristic in the sense that measurement always comes down in one way or another to the outcome of observing individuals' behavior, if nothing else, their verbal expressions, or marks on paper. Skinner (1974) objected to this methodological version of behaviorism when it treats behavior merely as a dependent variable that is an indicator of something other than behavior (e.g., mind, cognition, brain activity), hence encouraging the view that psychology is the science of behavior and mental life. The result is that behavior remains subsidiary to mentality or the nervous system. Skinner's (1938) often reiterated position that "behavior may be treated as a subject matter in its own right" (p. 440) does not question the importance of the subject matter of the neurosciences for behavior (Skinner, 1989, p. 130). He maintained that "we can predict and control behavior without knowing anything about what is happening inside [although] a complete account will nevertheless require the joint action of both sciences, each with its own instruments and methods" (Skinner, 1989, p. 130).

There are two major subtopics to consider in discussing Skinner's behavioral emphasis. The first concerns his definition of behavior, which is rather abstract. For example, to describe an episode of behavior, one cannot be restricted to the behaving organism. Skinner held that behavior is the action of the whole organism, not a piece of it: "It is the *organism as a whole* that behaves" (Skinner, 1975, p. 44). "Behavior is what an organism is doing" (Skinner, 1938, p. 6) that we can determine by observing its relations with its environment, that is, "the action of the organism upon the outside world" (Skinner, 1938, p. 6). Skinner's rejection of the activity of muscles and organs per se as the essence of psychological behavior is elaborated later in the section on Classification of Behavior Into Respondent and Operant.

The second important issue that arises from Skinner's thoroughgoing behavioral position concerns his apparent rejection of everything mental. The history of the behavioral movement involves confrontations between mentalistic and behavioristic investigative and explanatory attempts. Skinner viewed as misguided those behavioristic attempts that (a) ignore the events that serve as the controlling variables for talk of mind and consciousness, (b) substitute self-reports for the events referred to as mental, and (c) use behavior as an indicator of mental activity that is assumed to be fundamentally different from behavior. He distinguished public and private events, although he did not consider the two to differ in the stuff of which they are made (see Materialism section). Public events are those accessible to other observers, whereas private events are characterized by limited accessibility. That is, private events (e.g., a toothache) are observable only to the individual in whose body they are occurring. Because others cannot directly observe certain psychological events, observable behavioral (public) events are separated from so-called mental (private) events, thereby holding back a completely behavioral approach to the subject matter of psychology. Skinner, of course, was not embracing mental events per se but simply encouraging an analysis of those behavioral events that are described in mainstream psychology and in the vernacular as "mentalistic." Skinner (1963) suggested that "the problem of privacy may be approached in a fresh direction by

starting with behavior rather than with immediate experience" (p. 953) and that

> It is particularly important that a science of behavior face the problem of privacy. It may do so without abandoning the basic position of behaviorism. Science often talks about things it cannot see or measure. When a man tosses a penny into the air, it must be assumed that he tosses the earth beneath him downward. It is quite out of the question to see or measure the effect on the earth, but an effect must be assumed for the sake of a consistent account. An adequate science of behavior must consider events taking place within the skin of the organism, not as physiological mediators of behavior but as part of behavior itself. It can deal with these events without assuming that they have any special nature or must be known in any special way. The skin is not that important as a boundary. Private and public events have the same kinds of physical dimensions. (p. 953)

Thus Skinner encouraged investigation of so-called sensations, perceptions, images, thoughts, awareness, and the like but did not treat these private events as fundamentally different from any of the public organismic and environmental behavioral events that served as the original classes of events studied by behaviorism.

Reductionism and Nonreductionism

Skinner's position seems to be most ambiguous on the issue of reductionism, so much so that we offer two conflicting assumptions.

Reductionism: The Subject Matter of Psychology Is Reducible (at Least to Biology)

> Eventually, we may assume, the facts and principles of psychology will be reducible not only to physiology but through biochemistry and chemistry to physics and subatomic physics. (Skinner, 1947, p. 31)

> The behaving organism will eventually be described and explained by the anatomist and physiologist. As far as behavior is concerned, they will give us an account of the genetic endowment of the species and tell how that endowment changes during the lifetime of the individual and why, as a result, the individual then responds in a given way on a given occasion. (Skinner, 1975, p. 42)

> The physiologist of the future will tell us all that can be known about what is happening inside the behaving organism. His account will be an important advance over a behavioral analysis, because the latter is necessarily "historical"—that is to say, it is confined to functional relations showing temporal gaps. Something is done today which affects the behavior of an organism tomorrow. No matter how clearly that fact can be established, a step is missing, and we must wait for the physiologist to supply it. (Skinner, 1974, p. 215)

Nonreductionism: Behavior Cannot Be Completely Explained in Terms of Biology or Any Other "Lower-Level" Discipline

> [This work] is not necessarily mechanistic in the sense of reducing the phenomena of behavior ultimately to the movement of particles, since no such reduction is made or considered essential. (Skinner, 1938, p. 433)

> Behavior is an acceptable subject matter in its own right, and . . . it can be studied with acceptable methods and without an eye to reductive explanation. (Skinner, 1961, p. 64)

> We do not need an explicit account of the anatomy and physiology of genetic endowment in order to describe . . . behavior. . . . Nor do we need to consider anatomy and physiology in order to see how the behavior of the individual is changed by his exposure to contingencies of reinforcement during his lifetime and how as a result he behaves in a given way on a given occasion. (Skinner, 1975, pp. 42–43)

> A science of behavior will be needed for both theoretical and practical purposes even when the behaving organism is fully understood at another level. (Skinner, 1975, p. 43)

Discussion

The reductionism assumption states a reductionistic position whereby events on one level can be explained in terms of another supposedly more simple or basic level. However, statements such as those following the nonreductionism assumption argue against such a resolution of psychological events to biology or any other discipline and assert that behavior can and should be treated as a subject matter in its own right without appeal to another level of explanation.

If Skinner's position were unequivocally reductionistic, then it would be in agreement with conventional scientific materialism, according to which science deals with a material realm, as contrasted with a spiritual or idealistic one, and it is possible and desirable to attempt to explain events in terms of materialistic concepts that are at a lower level. The first quotation under Reductionism exhibits an extreme form of materialistic reductionism when subatomic physics is considered to be the eventual basis of explanation of psychological events.

Skinner exhibited a nonreductionistic side to his behaviorism, as the quotations under Nonreductionism show. At one point in *About Behaviorism,* he explicitly disavowed reductionism. He argued that his approach in no way advocated reducing any aspect of humans, stressing that behaviorism does not "*reduce* feelings to bodily states . . . *reduce* thought processes to behavior . . . or *reduce* morality to certain features of the social environment" (Skinner, 1974, p. 241).

Ambiguity is introduced into Skinner's seemingly antireductionistic position when, in referring to the physiologist of the future, he stated that "his account will be an important advance over a behavioral analysis, because the latter is necessarily 'historical'—that is to say, it is confined to functional relations showing temporal gaps" (Skinner, 1974, p. 215). Does this imply that someday psychology will be reduced to physiology? Evidence for such an implication seems to exist:

> In general I reject any appeal to physiology in explaining behavior on the grounds that physiology is still [as of the present] far less advanced than the analysis of behavior and has yet to [but may someday?] deal with the processes responsible for the behavior attributed to contingencies of reinforcement. (Skinner, 1982, p. 190)

It seems that the key to understanding Skinner's position on reductionism is that he endorsed a physiological analysis only insofar as it may prove useful in filling the temporal gaps in the functional analysis of the relations between the individual's exposure to causal environmental contingencies and ensuing behavior. Insofar as the purpose of behavior analysis is the prediction and control of behavior (see Purpose of Science), a functional analysis definitely has the upper hand over a physiological analysis. That the latter is a possibility, however, is seemingly not ruled out.

Organism as the Locus of Biological Change: The Organism Changes Through Evolutional and Environmental Histories, and the Changes are Biological

Evolutionary and environmental histories change an organism. (Skinner, 1971, pp. 195–196)

[The physiologist of the future] will be able to show how an organism is changed when exposed to contingencies of reinforcement and why the changed organism then behaves in a different way, possibly at a much later date. What he discovers cannot invalidate the laws of a science of behavior, but it will make the picture of human action more nearly complete. (Skinner, 1974, p. 215)

Both kinds of contingencies [phylogenic and ontogenic] change the organism so that it adjusts to its environment in the sense of behaving in it more effectively. (Skinner, 1966b, pp. 1211–1212)

People are changed by the contingencies of reinforcement, they do not store information about them. (Skinner, 1988e, p. 53)

Contingencies of reinforcement change the individual; as a result the individual now behaves in a different way. (Skinner, 1988d, p. 409)

The relation between environmental history and contemporary behavior is so important that we address this issue with a separate assumption. This assumption, concerning the locus and form of changes induced by selective consequences, clarifies why Skinner addressed biological–behavioral relations in the first place. It does not clarify where Skinner stood on reductionism–nonreductionism per se.

Above and beyond behavior, what changes when an organism is exposed to effective contingencies? Skinner's answer was that the organism changes but not in any conventionally psychological way (e.g., psychically, mentally, cognitively). In the case of natural selection, the organism has been changed by endowing it with a physiology that makes conditioning possible: "What has been selected appears to be a susceptibility to ontogenic contingencies" (Skinner, 1966b, p. 1208). In the case of operant conditioning, reinforcement contingencies change organisms biologically during their individual life spans.

Skinner's emphasis on selective contingencies changing organisms biologically counteracted explanations in terms of traditional psychological concepts of acquisition and storage devices. To restrict our attention to ontogenic contingencies, the explanatory problem is the temporal gap between exposure to reinforcement contingencies and subsequent behavioral change. Skinner viewed a biologically empty organism approach as unsatisfactory for handling the mediation of reinforcement effects over time, but, consistent with the fundamental assumption of materialism (see previous discussion), he found nonphysical (i.e., mentalistic) accounts equally lacking. He rejected the notion that the organism cognitively internalizes contingencies of reinforcement in such forms as information, knowledge, or expectations that require unknown storage mechanisms so they can be activated in the future on occasions when behavior occurs. The alternative to modified mental structures and processes with exposure to reinforcement contingencies is a changed organism, one changed biologically. To deny alterations of mental conditions is not to deny that reinforcement contingencies modify "what is felt as feelings or introspectively observed as states of mind . . . [for these] . . . are the products of certain contingencies of reinforcement" (Skinner, 1988b, p. 175).

Classification of Behavior Into Respondent and Operant: There Are Two Major Classes of Behavior or, More Completely, Functional Relations: Respondent and Operant

The kind of behavior that is correlated with specific eliciting stimuli may be called *respondent* behavior and a given correlation *a respondent*. The term is intended to carry the sense of a relation to a prior event. Such behavior as is not under this kind of control I shall call *operant* and any specific example *an operant*. (Skinner, 1938, p. 20)

[Two processes evolved] through which individual organisms acquired behavior appropriate to novel environments. Through respondent (Pavlovian) conditioning, responses prepared in advance by natural selection could come under control of new stimuli. Through operant conditioning, new responses could be strengthened ("reinforced") by events which immediately followed them. (Skinner, 1981, p. 501)

This assumption extends the definition of behavior brought up under Behavior as Subject Matter. The classification of behavior has its origins in the distinction between involuntary and voluntary behavior. According to Skinner, the term *involuntary,* when used properly, refers to elicited behavior or reflexes of the sort most associated with Pavlov's work. Mechanical causality applies to behavior of this class; the responses of stimulus–response functional relations are true responses insofar as they depend upon immediately prior events (Skinner, 1953, p. 64). The term *respondent* applies to this class of stimulus–response functional relations, and they are modified only in the sense that the eliciting stimuli of responses can be changed. Respondent conditioning refers to this process of changing eliciting stimuli. Thus, Skinner included work inspired by Pavlov in his behaviorism but found that class of behavior known in the vernacular as

"voluntary" behavior of much more interest and relevance for psychology.

Perhaps most of Skinner's novel contributions to behavioral science revolve around the operant class of behavior to which the new causal mode of selection by consequences (see Consequential Causality section) applies. Operant behavior is defined by functional relations between classes of responses (not specific instances of responses) and environmental consequences. The term *operant* "emphasizes the fact that the behavior *operates* upon the environment to generate consequences" (Skinner, 1953, p. 65).

A more complete definition of operant behavior requires a distinction between response instances and response classes (e.g., Skinner, 1953, p. 65, 1969, p. 131). The former is specifiable in terms of topography or structure and refers to a particular, specific occurrence of a response. For example, "The pigeon pecked a key at 1:30 p.m." and "Edwardo drank a glass of water between 4:00 and 4:02 p.m. today." On the other hand, key pecking, for example, regardless of when specific instances occur, defines a response class. It is a set of acts defined by a measurable impact on the environment (such as activation of a relay) that transcends particular instances and forms of response. Given this distinction between response instances and response classes, Skinner defined the operant as follows:

> The term emphasizes the fact that the behavior *operates* upon the environment to generate consequences. The consequences define the properties with respect to which responses are called similar. The term will be used both as an adjective (operant behavior) and as a noun to designate the behavior defined by a given consequence. (Skinner, 1953, p. 65)

According to this definition, an operant is identified with a response class that can be strengthened by events (reinforcers) that immediately follow it, but whether or not the operant is strengthened has no bearing on definition of the operant.

In theoretical, research, and practical applications of the operant construct, Skinner stressed contingencies between operants and consequential events as crucial for predicting and controlling operant behavior. In fact, Skinner (1969) also tied the definition of operant behavior to experimental demonstrations in which a response class is modified as a function of its consequences. For example,

> [In discussing the act of "flipping on a light switch":] The topography of the response is described accurately enough as "flipping the switch." *If the appearance of light is reinforcing* [italics added; "flipping the switch" becomes more likely] . . . the topography and the consequences define an operant. (p. 128)

> It is always a response [instance] upon which a given reinforcement [consequence] is contingent, but it is contingent upon properties which define membership in an operant. Thus a set of contingencies defines an operant. (p. 131)

> Allowing water to pass over one's hands can perhaps be adequately described as topography, but "washing one's hands" is an "operant" defined by the fact that, *when one has behaved this way in the past, one's hands have become clean—a condition which has become reinforcing* [italics added] because, say, it has minimized a threat of criticism or contagion. Behavior of precisely the same topography would be part of another operant if the reinforcement had consisted of simple stimulation (e.g., "tickling") of the hands. (p. 130)

Catania (1973) referred to the latter use of the term *operant* as a functional one ordinarily used in theoretical discussion. He noted that in the method section of experimental reports the term is used descriptively, that is, without requiring modifiability as a function of consequences.

Rate of Responding as Fundamental Datum

Skinner offered probability of response as the primary conceptual measure of his science. However, probability is not directly measurable, and measures such as latency and magnitude are not appropriate for operant behavior because such behavior is not elicited by antecedent stimuli. Skinner suggested that "in operant conditioning we 'strengthen' an operant in the sense of making a response more probable or, in actual fact, more frequent" (Skinner, 1953, p. 65). Thus, Skinner's solution to the unique character of the operant was to measure probability primarily by way of frequency of response, or, more precisely, "the length of time elapsing between a response [instance] and the response [instance] immediately preceding it or, in other words, the rate of responding" (Skinner, 1938, p. 58).

Purpose

The operant concept, in conjunction with several of the assumptions we have presented up to this point, is crucial for understanding Skinner's handling of the purpose, meaning, and intention of behavior. Consider an episode in which an individual "washed his or her hands." This response instance is part of history and is not an operant. However, what the individual was doing here may be properly regarded as operant behavior on the basis that when the person has behaved similarly in the past their hands were made cleaner, a condition that has become reinforcing because it has been socially praised, has reduced the likelihood of criticism, or has made contagion less likely. In this case, we may speak of the operant class of "hand washing." It is possible, as some of those who work with certain classes of individuals with disabilities will note, that the same topography participates in defining another operant such as "stimulating body" on the basis that the critical consequences of past instances have not been those that define hand washing but rather mere tactual stimulation. Intention and purpose are best understood in terms of controlling variables (see Methodology section):

> Purpose is not a property of behavior itself; it is a way of referring to controlling variables. (Skinner, 1953, p. 88)

> When someone says that he can see the meaning of a response, he means that he can infer some of the [potentially manipulable] variables of which the response is usually a function. (Skinner, 1957, p. 14)

Clean hands and body stimulation are very different consequences of performing the same topographies; the future orientations (purposes) of the two operants distinguish them.

The replacement of purpose with selection, in combination with consequential causality (see previous discussion), led Skinner to a proposal concerning the evolution of cultural practices. He fully discussed the implications of this (e.g., 1953, 1969, 1971, 1972, 1981, 1987a), beginning with the utopian novel *Walden Two* (1948). In our previous discussion of consequential causality, we pointed out that Skinner applied selection by consequences to phylogeny (biological natural selection) and to ontogeny (the behavior of individuals). At the level of the evolution of practices characteristic of a group of people, Skinner suggested that the selecting consequences are those that contribute to the survival of the group. Thus, cultural evolution is a third kind of natural selection. He argued that the delayed consequences of cultural practices (e.g., the controlled use of fire) are "too remote to reinforce the behavior of any member of the group" (Skinner, 1989, p. 117).

Stimulus Control of Operant Behavior: Operant Behavior Can Be Brought Under the Control of Antecedent Stimuli, and Description of Operant Behavior Usually Requires Three Elementary Terms and Their Functional Interrelations

> The occasion upon which behavior occurs, the behavior itself, and its consequences are interrelated in the contingencies of reinforcement. . . . As a result of its place in these contingencies, a stimulus present when a response is reinforced acquires some control over the response. It does not then elicit the response as in a reflex; it simply makes it more probable that it will occur again. (Skinner, 1974, pp. 73–74)

> An adequate formulation of the interaction between an organism and its environment must always specify three things: (1) the occasion upon which a response occurs, (2) the response itself, and (3) the reinforcing consequences. The interrelationships among them are the "contingencies of reinforcement." (Skinner, 1969, p. 7)

Under certain conditions, experimenters can reliably "turn on" and "turn off" operant behavior by presenting and removing stimuli. Such stimulus–response relations may even have the appearance of respondent relations; however, more detailed examination will reveal that these effects are systematic outcomes of the organism's previous exposure to response–consequence contingencies conditional on the presence or absence of particular stimuli. In the simplest case, occurrences of instances of a response class are followed by reinforcement only in the presence of a stimulus. This stimulus is said to acquire discriminative stimulus control over the operant. After discriminative stimulus control has developed, the experimenter can manipulate the stimulus as an independent variable to control the operant behavior. In this sense, the stimulus definitely is a controlling variable of the behavior in question. However, its status as a controlling variable is conditional inasmuch as functional relations between discriminative stimuli and behavior are dependent on a history of selective behavior–consequence relations (see Consequential Causality and Classification of Behavior Into Respondent and Operant sections) in the presence of the stimulus. In contrast to the eliciting function of stimuli in respondent relations, the discriminative stimuli of operant relations are said to "set the occasion" for responding. Even when discriminative stimuli have the surface characteristics of elicitors, because the history of their functional effects is traceable to operant behavior–consequence relations, it is not proper to fit them into the stimulus–response reflex causal mode.

To take into account the development of discriminative stimulus control of operants, description of behavior requires not only response–reinforcer functional relations but discriminative stimulus–response relations as well. The result is the "three-term" contingency—discriminative stimulus: response-reinforcing consequence. The three-term contingency forms the fundamental unit of analysis in the study of operant behavior.

Skinner's conception of the three-term contingency as the fundamental unit of stimulus control was a radical departure from the reflexological-derived stimulus–response model, according to which the organism could only respond (in the conventional sense of the term) to prior physical or mental stimuli. In terms of the operant framework, at the most elementary level, the crucially important class of behavior called *voluntary* is a matter of the selective altering of the probability of behavior by discriminative stimuli in the presence of which the behavior was selected by consequences in the history of the organism.

On the Generality of Behavioral Principles: The Full Complexities of Human Activity—Including Language, Thinking, Consciousness, and Science—Are Behaviors to Which All These Features Apply

This final assumption of Skinner's psychology reveals perhaps the most revolutionary portion of Skinner's careerlong attempt to permit no aspect of the human experience to remain untouched by scientific understanding. The breadth of his applications that this section reveals is further amplified by articles on the technology of education, psychotic behavior, artistic creativity, the genesis of a poem, literary products, and telepathy experiments (Skinner, 1972).

On Language

> Our first responsibility is simple *description:* what is the topography of this subdivision of human behavior? Once that question has been answered . . . we may advance to the stage called *explanation:* what conditions are related to the occurrence of the behavior—what are the variables of which it is a function? Once these have been identified, we can account for the dynamic

characteristics of verbal behavior within a framework appropriate to human behavior as a whole. (Skinner, 1957, p. 10)

Language has the character of a thing, something a person acquires and possesses. Psychologists speak of the "acquisition of language" in the child. The words and sentences of which a language is composed are said to be tools used to express meanings, thoughts, ideas . . . and many other things in or on the speaker's mind. A much more productive view is that verbal behavior is behavior. It has a special character only because it is reinforced by its effects on people—at first other people, but eventually the speaker himself. As a result, it is free of the spatial, temporal, and mechanical relations which prevail between operant behavior and nonsocial consequences. (Skinner, 1974, pp. 88–89)

Skinner (1945a, 1945b) addressed the four behavioral complexities included in this final assumption that affirm a thoroughgoing behavioral psychology. We obtain the first glimpse of an approach to language that is neither mentalistic nor in terms of stimulus–response behaviorism and one that Skinner subsequently used as a keystone to handle thought, what it means to be conscious, and the nature of science. Skinner (1957) rejected conventional views of language as the "use of words," the "communication of ideas," the "sharing of meaning," the "expression of thoughts," and so on. Consistent with his position on the subject matter of psychology (see Behavior as Subject Matter section), Skinner took language as behavior that is understandable in its own right. Given this, he found that (a) the term *verbal behavior* is preferable, (b) the behavior is of the voluntary (operant) class (see Classification of Behavior Into Respondent and Operant section), (c) the behavior is selected (see Consequential Causality) by environmental (see Locus of Behavioral Control) consequences, (d) the behavior is subject to functional analysis (see Methodology), and (e) the starting point for description is the three-term contingency (see Stimulus Control of Operant Behavior).

Although bearing a certain superficial similarity to nonhuman behavior under conditions of food-reinforced lever pressing or key pecking, verbal behavior operates in a fundamentally different way. We can describe in physical terms the nonhuman pressing a key or the human in a nonsocial situation who walks toward an object and picks it up. In contrast, humans frequently act "only indirectly upon the environment from which the ultimate consequences of [their] behavior emerge. . . . Instead of going to a drinking fountain, a thirsty man may simply 'ask for a glass of water' " (Skinner, 1957, p. 1). The special aspect of verbal behavior is that other persons are crucially involved, at least indirectly, in mediating the consequences of the speaker's behavior. As such, verbal behavior has special controlling variables (i.e., social ones). The role of others in controlling verbal behavior is favored over reference or correspondence theories of language. Skinner objected to the conventional view that words or collections of words stand for or refer to objects. Consistent with the discussion of purpose, intention, and meaning under Classification of Behavior Into Respondent and Operant, Skinner treated the meaning of words in terms of the variables determining their occurrence in any given instance. According to Skinner's treatment, we account for a remark and understand what it means by identifying the variables that control it. For example, we say that we see different meanings of "fire" when given as a command to a firing squad, when made in the presence of a burning building, and when preceded by "wind and rain."

Skinner's analysis of the role of verbal behavior led him to recognize a complexity in the nature of the relations controlling operants that took his behaviorism ever more fully into what psychologists typically regard as cognitive processes. Specifically, he proposed that operant behavior can be classified into two categories: contingency shaped and rule governed.

The response which satisfies a complex set of contingencies, and thus solves the problem, may come about as the result of direct shaping by the contingencies . . . or it may be evoked by contingency-related stimuli constructed either by the problem solver himself or by others. The difference between rule-following and contingency-shaped behavior is obvious when instances are pretty clearly only one or the other. (Skinner, 1966a, p. 241)

A person who is following directions, taking advice, heeding warnings, or obeying rules or laws does not behave precisely as one who has been directly exposed to the contingencies, because a description of the contingencies is never complete or exact . . . and because the supporting contingencies are seldom fully maintained. (Skinner, 1974, p. 125)

To this point, we have discussed one of the two classes of operant behavior. Contingency-shaped behavior is the class that most obviously (a) follows the principle of consequential selection and (b) comes under stimulus control and is described using the three-term contingency. Skinner (1966a) greatly extended the scope of his approach in the realm of human behavior with the addition of the rule-governed class of operant behavior. Rule-governed behavior occurs when the individual is behaving in accordance with explicit rules, advice, instructions, modeling performances, plans, maxims, and the like. Rules are contingency-specifying stimuli. Either directly or by implication from past experience, the rule specifies an environmental consequence of behaving in a certain way (e.g., "A free gift will be given to the first 100 individuals who enter the store" and "The student is required to attend all class meetings to pass this course").

If we advise someone how to "make friends and influence people," his or her behavior may change to a point at which he or she acts more in accordance with our rules. However, the ultimate effect of the advice rests on the environmental contingencies enjoined by our advisee's rule-following behavior. Rules function as discriminative stimuli (see Stimulus Control of Operant Behavior section), and a person will follow rules in the first place to the extent that "previous behavior in response to similar verbal stimuli [i.e., rules, advice] has been reinforced" (Skinner, 1966a, p. 244). Thus, consequential selection is central in the development and maintenance of rule-governed behavior albeit less directly so than with many

cases of straightforward contingency-controlled behavior. The pervasiveness of social factors in human behavior makes most cases of human behavior the product of both contingencies and rules.

On Thinking

> The simplest and most satisfactory view is that thought is simply *behavior*—verbal or nonverbal, covert or overt. It is not some mysterious process responsible for behavior but the very behavior itself in all the complexity of its controlling relations, with respect to both man the behaver and the environment in which he lives. (Skinner, 1957, p. 449)

> Mental life and the world in which it is lived are inventions. They have been invented on the analogy of external behavior occurring under external contingencies. Thinking is behaving. The mistake is in allocating the behavior to the mind. (Skinner, 1974, p. 104)

Implicit in Skinner's (1957) interpretation of verbal behavior is a listener who responds to the speaker's verbal stimuli. Those cases in which speakers also become listeners (e.g., talk to themselves), particularly when others cannot observe their behavior, describe (but not fully, see later discussion) the "special human achievement called 'thinking' " (Skinner, 1957, p. 433). This is not to suggest that Skinner took the early behavioristic position that identified thinking with subaudible talking. His view was considerably more sophisticated. The main characteristic of thinking is that persons behave with respect to themselves, which is to say that controlling relations do not involve other persons. Thus, the person's own behavior has an overt or covert self-stimulatory effect, once again illustrating that Skinner's framework for behavior was not formulated on the S-R, wind–weather vane model. Although thinking frequently is both covert (not readily observable by others) and verbal, it might be overt (if someone were in the presence of the behaver, he or she could readily observe it), and it is not restricted to verbal behavior. The critical mark of thinking is not tied to distinctions between verbal and nonverbal, overt and covert, and private and public nor to weak and strong behavior (Skinner, 1957, 1989). According to Skinner, thinking is behaving, either verbal or nonverbal, overt or covert, weakly or strongly.

In our view, one of the most subtle and pregnant features of Skinner's behaviorism concerns the assumption that thinking can be covert or overt nonverbal behavior. Recall that the definition of verbal behavior requires a special character: the participation of social consequences that may include speakers themselves. Nonverbal behavior entails no participating social consequences; the consequences are components of the behaver's nonsocial environment. What this comes down to is that

> To think is to do something that makes other behavior possible. Solving a problem is an example. A problem is a situation that does not evoke an effective response; we solve it by changing the situation until a response occurs. Telephoning a friend is a problem if we do not know the number, and we solve it by looking up the number. (Skinner, 1989, p. 20)

To us, it seems that Skinner took the radical position that all thinking is operant behavior and all operant behavior is thinking. This appears to be the message "behind" Skinner's (1957) assertion that "so far as a science of behavior is concerned, Man Thinking is simply Man Behaving" (p. 452) and in remarks such as "in the broadest possible sense, the thought of Julius Caesar was simply the sum total of his responses to the complex world in which he lived" (Skinner, 1957, pp. 451–452) and "cognitive processes are behavioral processes; they are things people do" (Skinner, 1989, p. 23). If we are correct, operant behavior—Skinner's foremost subject throughout his career—completely encompasses one of the main strongholds of the mentalistic psychology that he aspired to replace with his behaviorism.

On Consciousness

> Being conscious, as a form of reacting to one's own behavior, is a social product. . . . The individual becomes aware of what he is doing only after society has reinforced verbal responses with respect to his behavior as the source of discriminative stimuli. The behavior to be described (the behavior of which one is to be aware) may later recede to the covert level, and (to add to a crowning difficulty) so may the verbal response. It is an ironic twist, considering the history of the behavioristic revolution, that as we develop a more effective vocabulary for the analysis of behavior we also enlarge the possibilities of awareness, so defined. (Skinner, 1945a, p. 277)

> I believe that all nonhuman species are conscious in the sense [that they see, hear, feel, and so on], as were all humans prior to the acquisition of verbal behavior. . . . But they do not observe that they are doing so. . . . A verbal community asks the individual such questions as, "What are you doing?," "Do you see that?," "What are you going to do?," and so on and thus supplies the contingencies for the self-descriptive behavior that is at the heart of a different kind of awareness or consciousness. (Skinner, 1988a, p. 306)

The previous discussion of Skinner's position on behavior as the subject matter of psychology (see Behavior as Subject Matter) indicates that he did not avoid private events. His handling of thinking is an example of how he forthrightly confronted the problem of privacy. Consciousness or awareness is another aspect of human experience that is often taken as private. Skinner interpreted consciousness such that it is not restricted to humans but yet he retained the uniqueness of human consciousness. Far from treating humans as but complicated rats or pigeons, Skinner argued that there is something special about human behavior.

Skinner (1974) distinguished between two classes of consciousness. Consciousness$_1$ (our designation) pertains to organisms' "awareness" of their environment, as when we say they are "conscious of their surroundings." Persons who have been rendered unconscious are no longer under the stimulus control of events either inside the body or outside it, and they may talk without being "conscious of their effects on listeners." Humans and nonhumans alike are conscious in this respect.

They feel pain in the sense of responding to painful stimuli, as they see a light or hear a sound in the sense of responding appropriately, but no verbal contingencies make them conscious [consciousness$_2$] of pain in the sense of feeling that they are feeling, or of light or sound in the sense of seeing that they are seeing or hearing that they are hearing. (Skinner, 1974, p. 220)

Thus, for Skinner, consciousness$_2$ was (probably) restricted to humans because of its social–verbal nature. Other persons arrange verbal contingencies for behavior descriptive of our behavior, permitting us to state rules about it and its relation to controlling variables. In tying together the uniquely human consciousness$_2$ with verbal contingencies that are always social, Skinner's behaviorism has a "radically" social cast that does not seem to be much appreciated in conventional social psychology.

On Science

The behavior of the logician, mathematician, and scientist is the most difficult part of the field of human behavior and possibly the most subtle and complex phenomenon ever submitted to a logical, mathematical, or scientific analysis, but because it has not yet been well analyzed, we should not conclude that it is a different kind of field, to be approached only with a different kind of analysis. (Skinner, 1974, p. 235)

Scientific knowledge is verbal behavior, though not necessarily linguistic. It is a corpus of rules for effective action, and there is a special sense in which it could be "true" if it yields the most effective action possible. (Skinner, 1974, p. 235)

The scientist first interacts with the world, like everyone else, in contingency-shaped behavior. He becomes a scientist when he begins to describe the contingencies and to design experiments which make them clearer. The ultimate product, the "laws" of science, governs scientific behavior as a corpus of rules to be followed. The behavior of the scientist in following them is reinforced by the same consequences as the original contingency-shaped behavior, but the controlling stimuli are different. (Skinner, 1988c, p. 197)

In the article "The Operational Analysis of Psychological Terms," Skinner (1945a) took the standpoint of functional analysis (see Methodology section) in his approach to the question of terms, concepts, and constructs. Basically, his position was that because the knowing profession of science offered as products verbal behavior, the preferred approach to epistemology was to describe the behavior of the scientist. Thus, a scientific term or knowledge claim has the same status as any other piece of human behavior. He later elaborated on the implications of an empirical epistemology (e.g., Skinner, 1957, 1963, 1974). According to this way of approaching knowledge, terms are not important. What is important are the conditions under which the scientist uses the term (i.e., the controlling variables). In other words, the meaning of scientific terms is handled as is the meaning of any other behavioral episode (see Classification of Behavior Into Respondent and Operant). Terms and knowledge claims do not mean anything more than their controlling variables. They do not convey otherwise unknowable information about things.

Skinner's radically naturalistic approach to epistemology has received little attention to date, and we doubt that few but the most devoted students of behaviorism are familiar with it. However, some writers have begun their own extensions of Skinner's epistemology to issues in the philosophy of science (Hineline, 1980; Lamal, 1983; Malagodi, 1986; Malagodi & Jackson, 1989; Schnaitter, 1978, 1984; Williams, 1986; Zuriff, 1980, 1985). As workers continue to take an operant standpoint to delve into behavioral complexities (e.g., those associated with the distinction between contingency-shaped and rule-governed behavior), it seems likely that we will hear more of this behavioral–psychological interpretation of what it means to know.

Eventually we shall be able to include, and perhaps to understand, our own verbal behavior as scientists. If it turns out that our final view of verbal behavior invalidates our scientific structure from the point of view of logic and truth-value, then so much the worse for logic, which will also have been embraced by our analysis. (Skinner, 1945a, p. 277)

Conclusion

The purpose of the present research was to examine the assumptive base of Skinner's behaviorism. This revealed a coherent set of fundamental assumptions. For those interested in deepening their understanding of Skinner's complex and influential approach to psychology, we recommend that they keep these assumptions available while they examine authoritative presentations such as those of Catania (1980), Day (1975, 1983), Michael (1985), and Reese (1986) and, above all, Skinner's writings themselves (e.g., 1953, 1957, 1974).

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